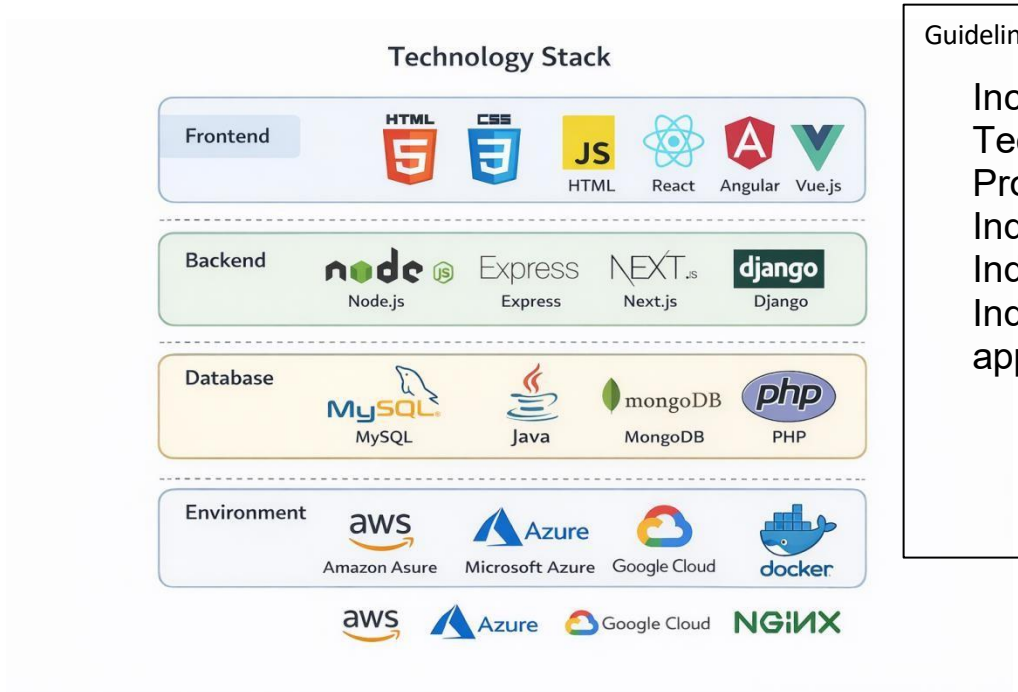
Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	20 february 2026
Team ID	LTVIP2026TMIDS83621
Project Name	Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Users interact with the system through Web and Mobile interface for registration, login, browsing heritage sites, booking tickets, and feedback submission. Admin and Executive dashboards are also provided.	HTML, CSS, JavaScript, ReactJS / Angular (Frontend), Bootstrap
2.	Application Logic-1	Handles user authentication including registration, login, OTP verification, password reset, and role-based access control.	Java (Spring Boot) / Python (Django / Flask)
3.	Application Logic-2	Manages heritage site operations such as add, update, delete, search, filter, and display detailed site information.	Java Spring Boot / Python Django REST API
4.	Application Logic-3	Handles booking management, ticket calculation, payment processing, feedback management, and report generation.	Java / Python, REST APIs
5.	Database	Stores user data, heritage site details, bookings, payments, feedback, and admin records.	MySQL / PostgreSQL
6.	Cloud Database	Cloud-based database service for scalability, backup, and high availability.	AWS RDS / Firebase / IBM DB2 on Cloud
7.	File Storage	Stores heritage images, booking receipts (PDF), and related documents	AWS S3 / Cloud Storage / Local File System
8.	External API-1	Used for payment gateway integration and email/OTP	Razorpay API / Stripe API / Twilio API / SMTP
9.	External API-2	Used for location services and map integration to show heritage site location.	Google Maps API
10.	Machine Learning Model	Used for location services and map integration to show heritage site location	Google Maps API
11.	Infrastructure (Server / Cloud)	Used to recommend popular heritage sites or analyze user preferences (optional enhancement).	Python (Scikit-learn) / TensorFlow

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	The system uses open-source frameworks for frontend and backend development to ensure flexibility, cost- effectiveness, and maintainability. Frontend framework is used for responsive UI. Backend framework handles REST APIs, business logic, and authentication.	ReactJS / Angular (Frontend), Spring Boot (Java) / Django (Python) (Backend)
2.	Security Implementations	The system implements secure authentication and authorization mechanisms. Passwords are encrypted before storing in the database. Role-Based Access Control (Admin, Customer, Executive, Customer Care) is implemented. HTTPS protocol is used for secure communication. Protection against SQL Injection, XSS, and CSRF attacks is ensured.	SHA-256 / BCrypt Password Encryption, JWT Authentication, HTTPS (SSL/TLS), Spring Security / Django Security, OWASP Guidelines
3.	Scalable Architecture	The system follows a 3-tier architecture (Presentation Layer, Application Layer, Database Layer). It supports horizontal and vertical scaling. Load balancing and cloud deployment enable handling multiple concurrent users. Microservices architecture can be adopted for future expansion.	3-Tier Architecture, REST APIs, Docker (Containerization), Kubernetes, AWS EC2 / Azure
4.	Availability	The system ensures high availability with minimal downtime. Cloud hosting enables 24/7 accessibility. Automated database backup and failover mechanisms are implemented. Load balancers distribute traffic efficiently.	AWS Cloud / Azure Cloud, Load Balancer, Cloud Auto Scaling, RDS Backup Services
5.	Performance	The system is optimized for fast page loading and quick booking confirmation. Database indexing improves query speed. Caching is implemented for frequently accessed heritage site data. CDN is used for faster image loading. The system supports multiple concurrent users efficiently.	MySQL Indexing, Redis Cache, CDN (Cloudflare), Optimized REST APIs